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Experiment 02

Aim: To Containerize Application using Docker

This experiment maps to LO3:

LO4: Containerization, orchestrate and deploy applications using Docker, Compose, Kubernetes and Cloud infrastructure tools.

Theory:

Docker is an open-source platform used to develop, package, and deploy applications in lightweight, portable units called **containers**. It allows applications to run consistently across different environments by including all the required dependencies and configurations.

Containerization is the process of packaging an application along with its libraries, runtime, and dependencies into a Docker container. This ensures the application behaves the same on a developer's system, a testing environment, or a production server.

A **Docker image** is a read-only template that contains the application code, runtime, libraries, and configuration files required to run an application. Images are created using a **Dockerfile** and act as the blueprint for containers.

A **Docker container** is a running instance of a Docker image. It provides an isolated environment where the application executes without affecting other applications or the host system. Multiple containers can be created from a single Docker image.

Difference Between Docker Image and Docker Container

Docker Image	**Docker Container**
Read-only template	Running instance of an image
Used to create containers	Used to run applications

Docker Image	Docker Container
Created using a Dockerfile	Created using the docker run command
Cannot execute on its own	Executes the application

Advantages of Docker

- Ensures application portability across different systems.
- Provides isolated environments for applications.
- Simplifies deployment and dependency management.
- Uses fewer system resources than virtual machines.
- Enables faster development, testing, and deployment.

1. Docker Installation Steps

    1. Download Docker Desktop from the official Docker website: https://www.docker.com/products/docker-desktop/

    2. Run the installer and follow the installation wizard.

    3. Restart your computer if prompted.

    4. Open Docker Desktop and wait until it shows Docker is running.

    5. Open Command Prompt or PowerShell and verify the installation:

2. docker --version

    6. Check Docker information:

3. docker info

7. Test the installation by running the Hello World container:

4. docker run hello-world

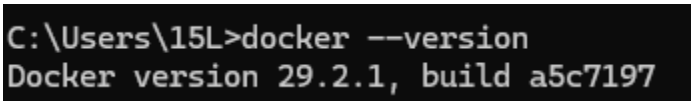
Result

Docker Desktop was installed successfully, and the installation was verified using the docker --version and docker run hello-world commands.

Commands Executed:

1. Check Docker Version

docker –version



2. Search for Docker Images

docker search ubuntu

This command searches Docker Hub for images related to **Ubuntu**.

```
C:\Users\15L>docker search ubuntu
NAME                                DESCRIPTION                                     STARS     OFFICIAL
ubuntu                              Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                        Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                        BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                        Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                    ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                      Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                   Prometheus is a systems and service monitori…   80
ubuntu/jre                          Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet                Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                       A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                        MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                     PostgreSQL is an open source object-relation…   45
ubuntu/redis                        Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime               Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                  Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                      Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                    Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                          Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                     Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                        Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                         Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                      Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector      This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager                 Ubuntu Rock for Alertmanager. It handles ale…   1
```

3. Search with a Filter

docker search ubuntu --filter stars=100

This command displays Ubuntu images that have **100 or more stars** on Docker Hub.

```
C:\Users\15L>docker search ubuntu --filter stars=100
NAME              DESCRIPTION                                     STARS     OFFICIAL
ubuntu            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2    Apache, a secure & extensible open-source HT…   104
```

4. Download an Ubuntu Image

docker pull ubuntu

This command downloads the latest Ubuntu image from Docker Hub to the local system.

```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```

5. List Downloaded Images

docker images

```
C:\Users\15L>docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       25.9kB         9.52kB    U
nginx:latest         5a88c9c45479        241MB           66MB    U
redis:latest         1c054d54ecd1        204MB         55.3MB    U
ubuntu:latest        3131b4cc82a7        161MB         45.3MB    U
```

6. Run Ubuntu Container

docker run -it ubuntu bash

```
C:\Users\15L>docker run -it ubuntu bash
root@654f898394b6:/#
```

7. Check Ubuntu Version (Inside Container)

cat /etc/os-release

```
C:\Users\15L>docker run -it ubuntu bash
root@654f898394b6:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

8. Exit the Container

exit

```
root@654f898394b6:/# exit
exit
```

9. List All Containers

docker ps -a

```
C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS                          PORTS
          NAMES
654f898394b6   ubuntu         "bash"                   About a minute ago   Exited (0) 34 seconds ago
          vigilant_albattani
f7b09cfcb56e   nginx          "/docker-entrypoint.…"   18 hours ago         Created
          practice
e41f9ed60c3c   nginx          "/docker-entrypoint.…"   18 hours ago         Exited (255) 20 minutes ago   0.0.0.0:8080->
80/tcp   mynginx
1000bc0dcc79   ubuntu         "/bin/bash"              19 hours ago         Exited (0) 19 hours ago
          mystifying_wilson
b00193fd8c8f   ubuntu         "/bin/bash"              19 hours ago         Exited (0) 19 hours ago
          jolly_hofstadter
ce280af76907   ubuntu         "/bin/bash"              19 hours ago         Exited (0) 19 hours ago
          sharp_pasteur
04ff3ca987f1   hello-world    "/hello"                 4 months ago         Exited (0) 4 months ago
          focused_agnesi
506fbcc93880   redis:latest   "docker-entrypoint.s…"   4 months ago         Exited (255) 19 hours ago     6379/tcp
          distracted_babbage
d3e8d53b8e82   redis          "docker-entrypoint.s…"   4 months ago         Exited (255) 19 hours ago     6379/tcp
          competent_rosalind
```

10. Start the Stopped Container

First note the **Container ID** from docker ps -a.

docker start <container_id>

```
C:\Users\15L>docker start 04ff3ca987f1
04ff3ca987f1
```

11. Stop the Running Container

docker stop <container_id>

```
C:\Users\15L>docker stop 04ff3ca987f1
04ff3ca987f1
```

12. Inspect a Container

docker inspect <container_id>


```
C:\Users\15L>docker inspect 04ff3ca987f1
[
    {
        "Id": "04ff3ca987f15183d1ee4e0fe422c86e8e26e9319af2cc760921bfd90717d43e",
        "Created": "2026-03-05T09:03:37.015302189Z",
        "Path": "/hello",
        "Args": [],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-21T05:07:31.720720878Z",
            "FinishedAt": "2026-07-21T05:07:31.807310959Z"
        },
        "Image": "sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a",
        "ResolvConfPath": "/var/lib/docker/containers/04ff3ca987f15183d1ee4e0fe422c86e8e26e9319af2cc760921bfd90717d43e/r
```

13. View Running Containers

docker ps

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

14. Remove a Container

docker rm <container_id>

```
C:\Users\15L>docker rm 04ff3ca987f1
04ff3ca987f1
```

15. Remove the Ubuntu Image

docker rmi ubuntu

```
C:\Users\15L>docker rmi ubuntu
Untagged: ubuntu:latest
Deleted: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
```

Conclusion:

I learned how to use Docker commands for container management, including checking Docker installation, searching and downloading images from Docker Hub, creating and running containers, viewing container details, and managing images. I also learned commands like docker pull, docker images, docker run, docker ps, docker stop, docker rm, and docker rmi for handling Docker images

and containers. This experiment helped me understand the basics of Docker containerization and how Docker is used for efficient application deployment, isolation, and environment management..